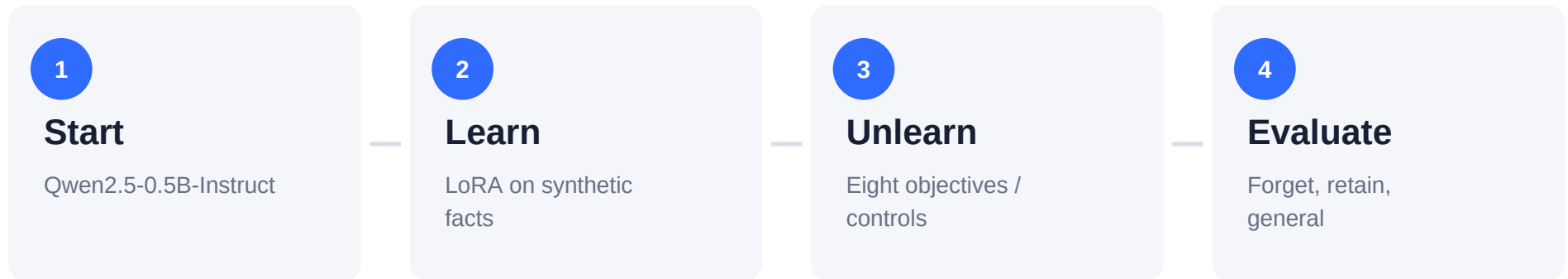


Selective forgetting in a small language model

Progress, trade-offs, negative results, and the next decision

Can a model forget selected facts without losing useful behavior?



Success requires low forget accuracy and high retain/general accuracy at the same time.

The controlled benchmark separates memorization from preservation

100

synthetic personas

20 / 80

forget / retain
personas

5

fact categories
per persona

500 / 1000

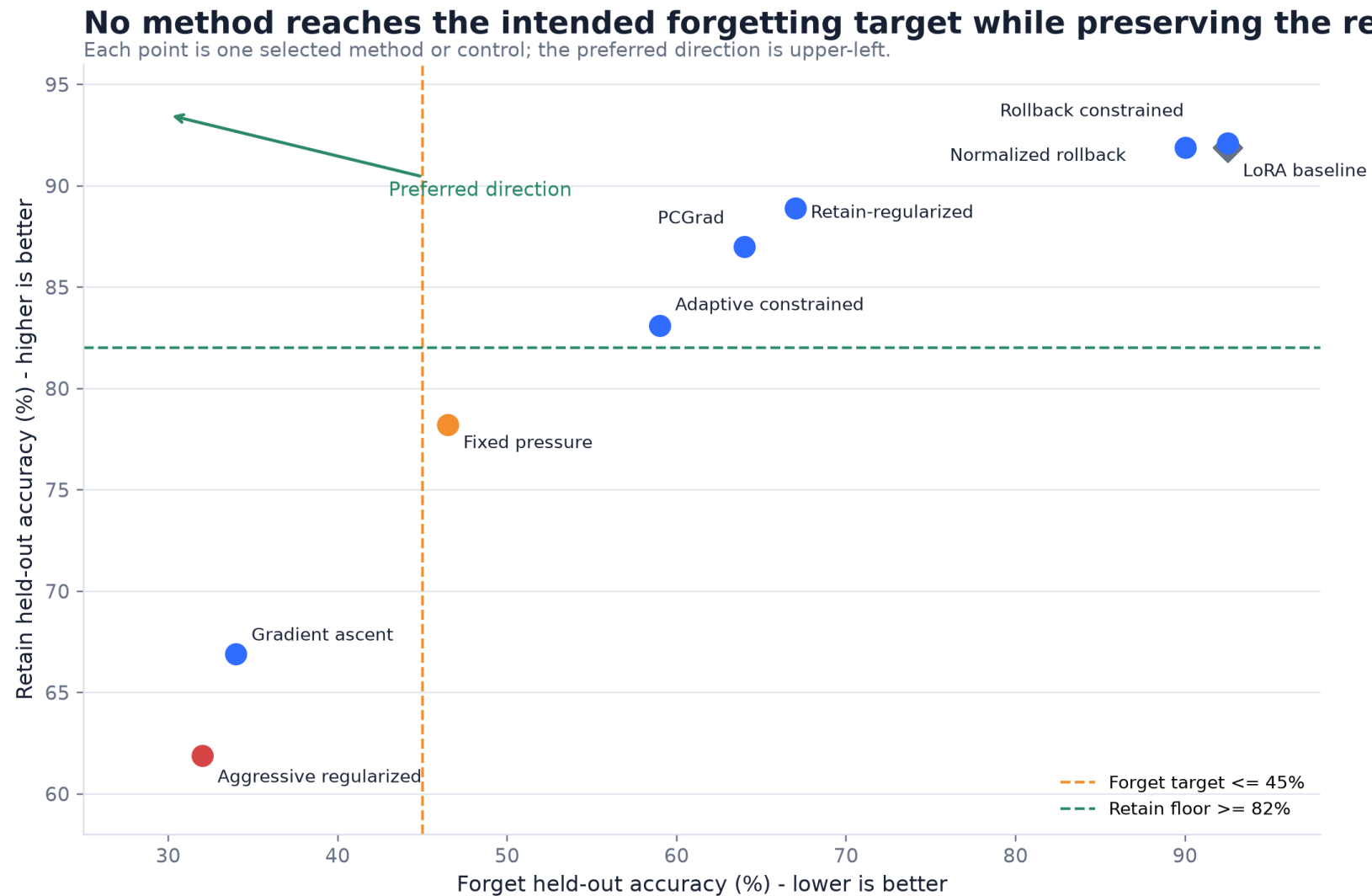
seen / held-out
questions

50

general-control
questions

Held-out questions paraphrase the same facts. They test prompt generalization, not whether the information is irrecoverable under adversarial extraction.

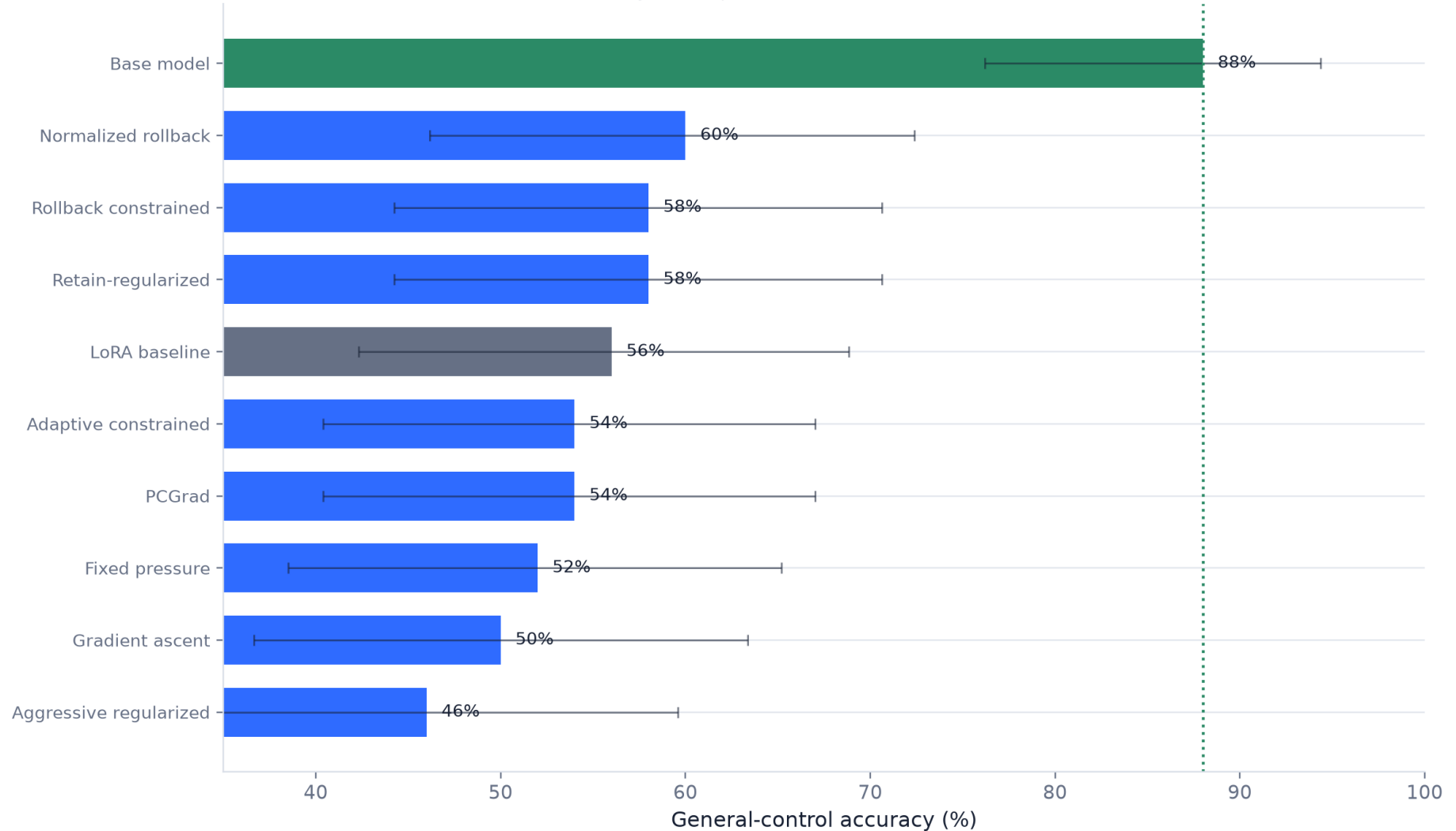
No method reaches both the forgetting target and retain floor



The largest utility loss happens before unlearning

Synthetic-fact LoRA causes the largest observed general-control drop

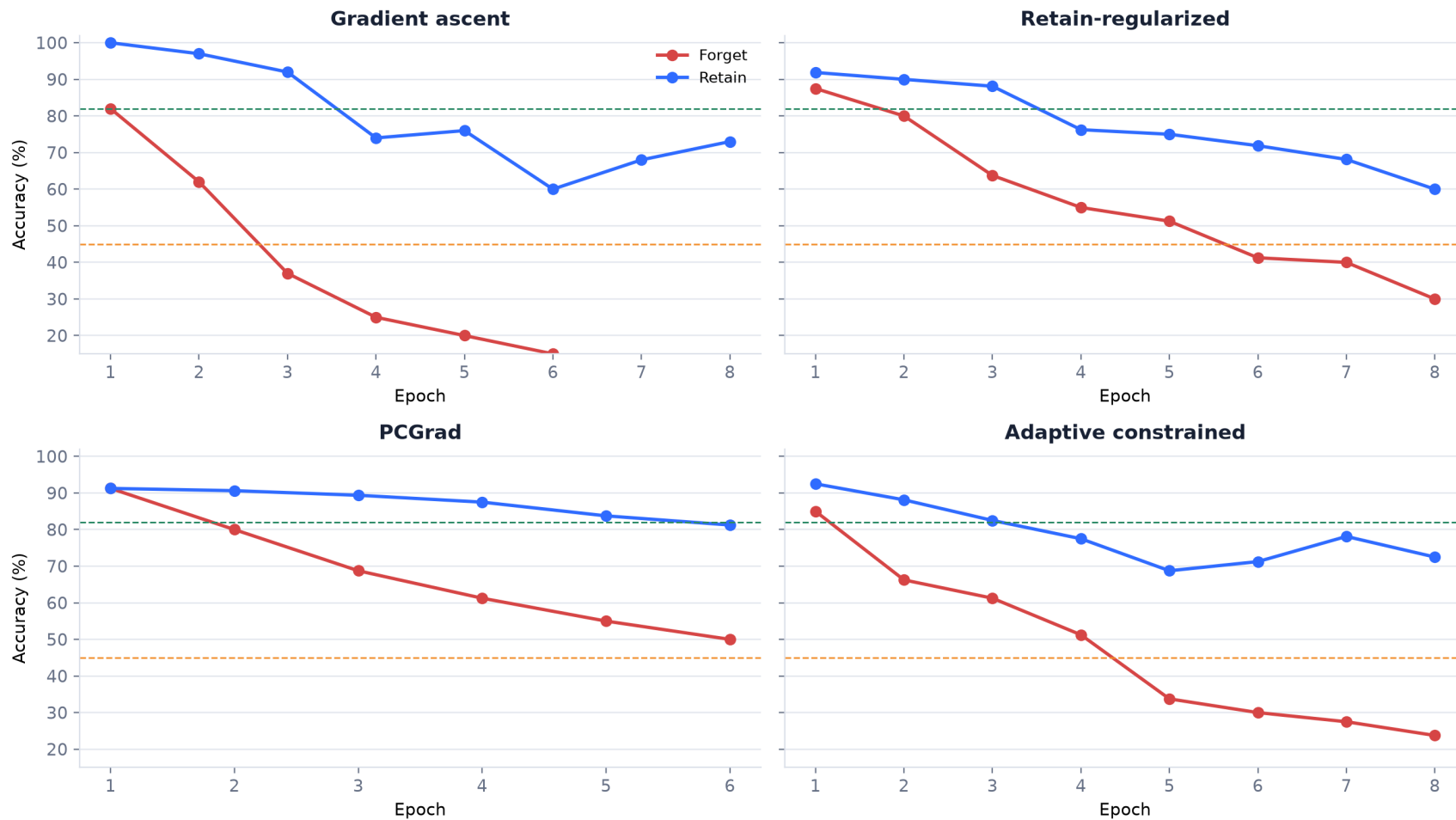
Whiskers are Wilson 95% intervals over only n=50 questions; small method differences are uncertain.



Training longer strengthens forgetting and eventually breaks retention

Training trajectories show where forgetting begins to damage retention

Week 4 uses train/sample accuracy; later weeks use held-out selection subsets.



Accuracy changes correspond to visibly different model behavior

Gradient ascent

Prompt: Which lab number is assigned to Parsa Aftab?

Before: E-254 | After: 426

Rollback constrained

Prompt: Which city does Sina Tavakol like most?

Before: Yazd | After: Yazd

The full qualitative table is exported as CSV. These examples illustrate behavior; they do not replace extraction-oriented privacy tests.

The next milestone is stronger evidence, not another optimizer

1. Train a retain-only oracle baseline.
2. Repeat key methods across seeds.
3. Add probability and extraction tests.
4. Diagnose the LoRA general-control drop.
5. Scale to 1.5B after the protocol stabilizes.

Decisions requested from the professor

- What retain threshold defines success?
- Oracle and seeds first, or scale the model now?
- What evidence is sufficient to claim removal?